ST.JOSEPH'S
COLLEGE OF ENGINEERING
AND TECHNOLOGY,
- PALAI -

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSD415 PROJECT PHASE 2

Title: **SMART ECO**

Date :13 May 2024

Team and Guide Details

- Team No. 08
- Team Members: Alen Mathew Tom
  Christi Joseph
  Mathews P Mathew
  Naveen S Pananthanam
- Project Guide: Prof. Maria Yesudas

COURSE OUTCOMES

CO1	Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: **Apply**).
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: **Apply**).
CO3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: **Apply**).
CO4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: **Apply**).
CO5	Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: **Analyze**).
CO6	Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level:**Apply**).

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ABSTRACT

SmartEco, Smart Home Energy Management System (SHEMS) pioneers IoT innovation by employing advanced current sensors to monitor and optimize residential energy consumption. Its Firebase-based backend ensures robust data storage, analysis, and security, with seamless authentication for user privacy. Unique predictive analytics forecasts billing estimates and encourages energy-saving habits through motion detection alerts, while the user-friendly Flutter-based mobile app provides effortless control and access to energy usage data. SmartEco simplifies energy management, empowering users to embrace sustainability and cost-effectiveness seamlessly.

INTRODUCTION

1. Integration of IoT innovation: SHEMS incorporates advanced IoT technology, utilizing current sensors to monitor energy consumption and identify energy-intensive appliances, thereby optimizing energy usage.

2.Predictive analytics: SHEMS goes beyond simple consumption monitoring by employing predictive analytics to forecast billing estimates, enabling users to take proactive measures in managing their energy costs.

3.User-friendly experience: The Flutter-based mobile application provides an intuitive interface for users to access real-time energy usage data and control the system effortlessly, enhancing user experience and making energy management more accessible.

PROBLEM DEFINITION

"The absence of tools for monitoring and optimizing energy usage contributes to the development of inefficient consumption patterns, exacerbated by rising electricity costs. Handheld device-based remote control of electronic devices offers convenience and potential energy savings. Addressing the growing financial burden on households necessitates empowering users with efficient energy management solutions. "

Journal Name	Outcome
That ‘internet of things’ thing,” *RFID journal .*	The authors highlight the transformative potential of IoT in connecting everyday objects to the internet, enabling new applications and services across various industries.
A vision, architectural elements, and future directions.	In this it discusses how this revolutionize by connecting and enabling objects to communicate and exchange data, leading to increased efficiency and innovation.
Research on the architecture and key technology of Internet of Things (IoT).	The authors investigate how IoT can enhance the efficiency and sustainability of smart grids by enabling real-time monitoring and control of energy distribution.
Controller and scheduler for smart homes, Sustainable Computing: Informatics and Systems.	This research aims to minimize energy wastage, reduce electricity costs, and enhance overall efficiency in smart home environments.
A smart home energy management system using IoT and big data analytics approach.	It helps IoT devices for real-time monitoring and control and utilizing big data analytics for data-driven insights
An efficient power scheduling scheme for residential load management in smart homes.	By introducing an efficient scheduling scheme that considers factors such as time-of-use pricing and user preferences, the system aims to improve energy efficiency, reduce costs, and promote sustainability

SRS

FUNCTIONAL REQUIREMENTS

HARDWARE Requirements

- AC/DC Power Module
- ESP8266 Microcontroller
- RCWL-0516 Motion Sensor
- ACS712 Current Sensor
- ATMEGA1608 Microcontroller
- Relay Module
- PCB Board
- Connecting Wires
- Resistors
- Capacitors

SOFTWARE Requirements

- Arduino IDE
- Fusion 360
- Flutter
- Firebase

TECHNOLOGY STACK

Figma

Firebase

Non-FUNCTIONAL REQUIREMENTS

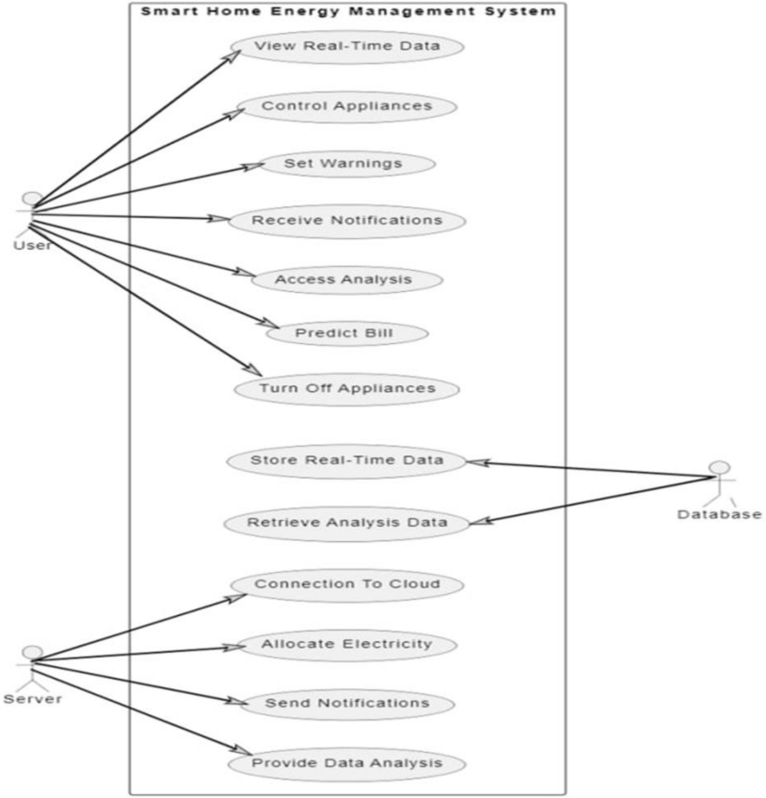
- Reliability
- Performance
- Scalability
- Security

PROPOSED SYSTEM

Our project presents a comprehensive smart home energy management solution comprising hardware components such as socket pins, relays, and current sensors, integrated with a mobile application developed using Flutter. This system enables users to remotely control up to three devices, monitor real-time power consumption, and receive alerts for motion detection. By synchronously managing devices via physical switches and the mobile app, users can optimize energy usage, while motion-based alerts enhance both energy efficiency and security, making it a versatile and user-friendly solution for modern smart homes.

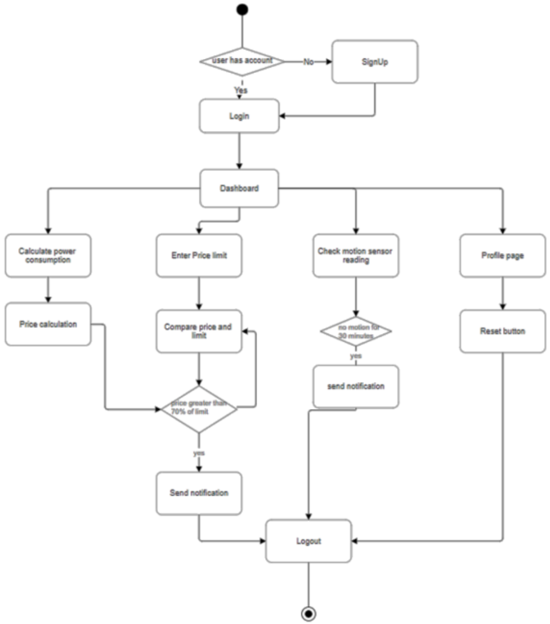
DESIGN

Use-case Diagram



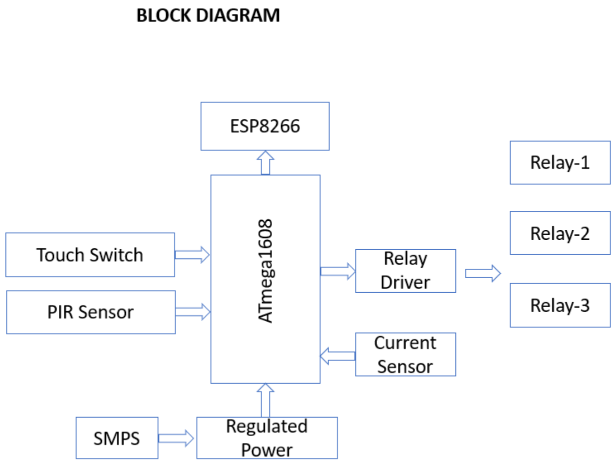
DESIGN

Activity Diagram

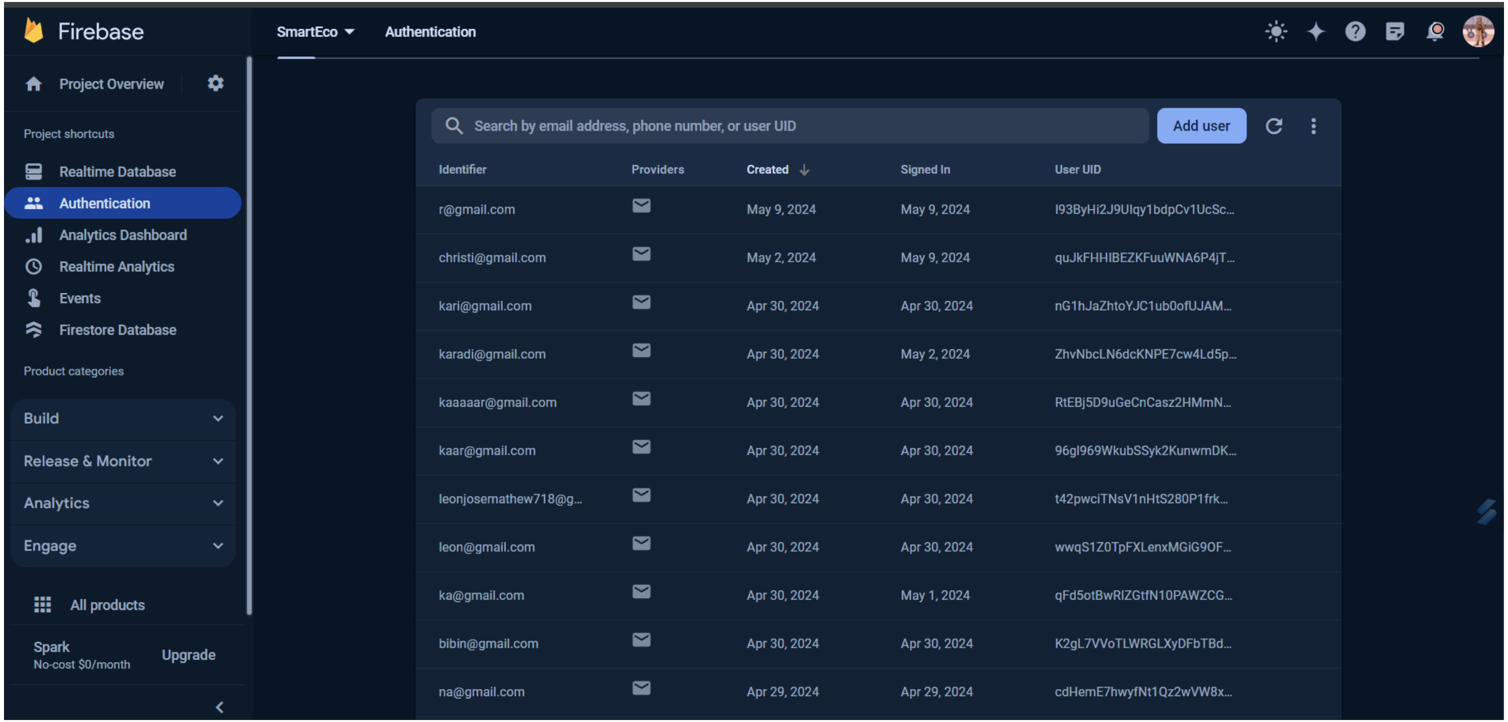


DESIGN

Block Diagram



RESULTS


Firebase
SmartEco
Authentication
Project Overview
Project shortcuts
Realtime Database
Authentication
Analytics Dashboard
Realtime Analytics
Events
Firestore Database
Product categories
Build
Release & Monitor
Analytics
Engage
All products
Spark
No-cost $0/month
Upgrade
Search by email address, phone number, or user UID
Add user
Identifier
Providers
Created
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User UID
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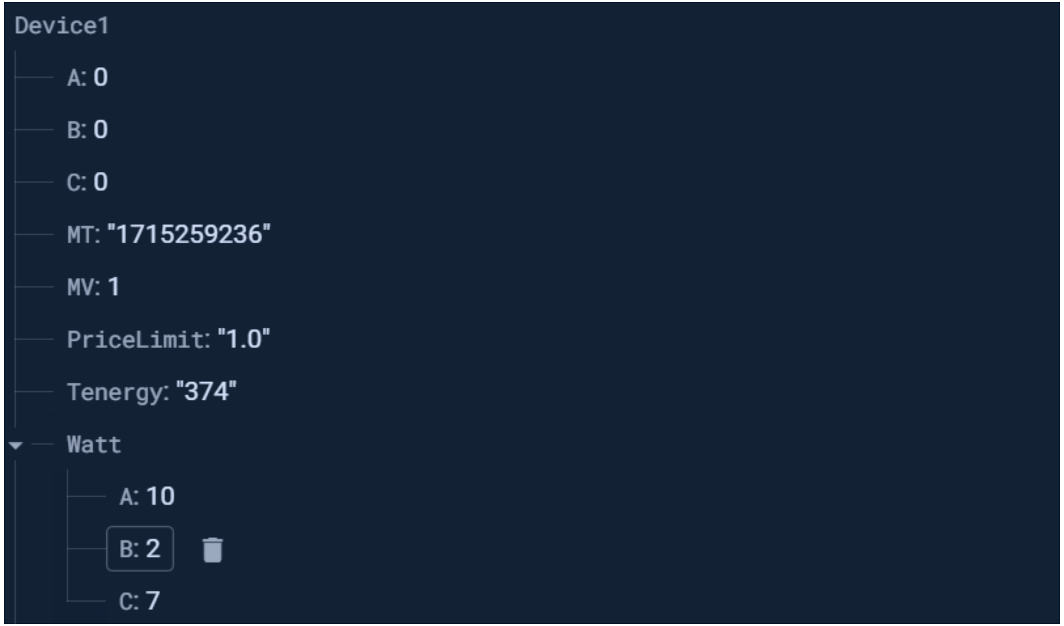
RESULTS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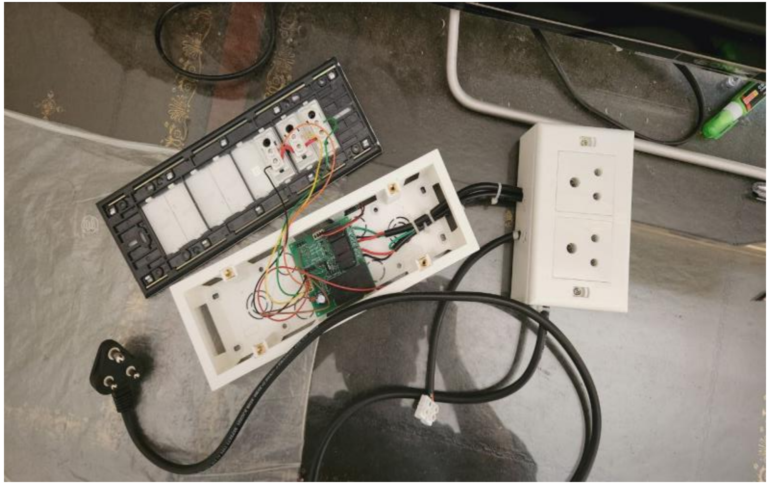
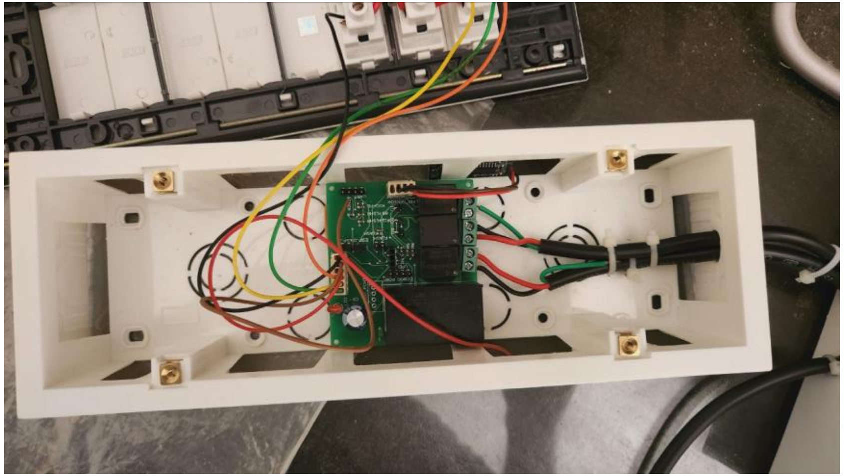
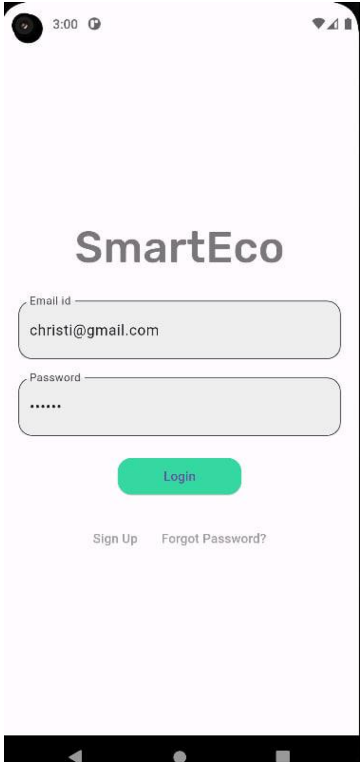
A: 0
B: 0
C: 0
MT: "1715259236"
MV: 1
PriceLimit: "1.0"
Tenergy: "374"
Watt
A: 10
B: 2
C: 7

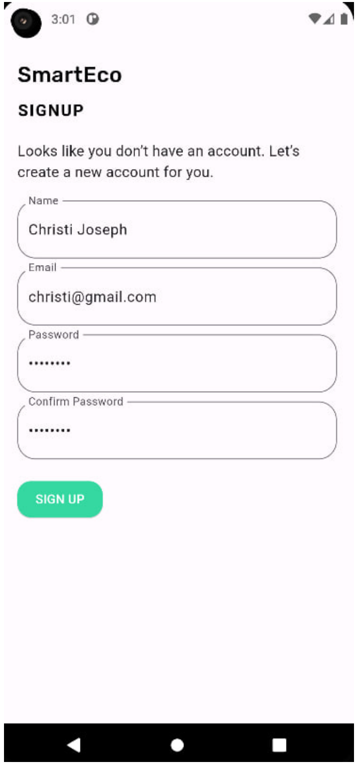
HARDWARE





APPLICATION UI





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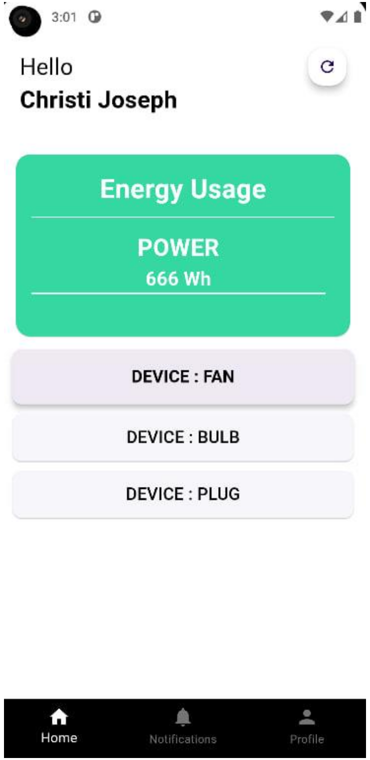

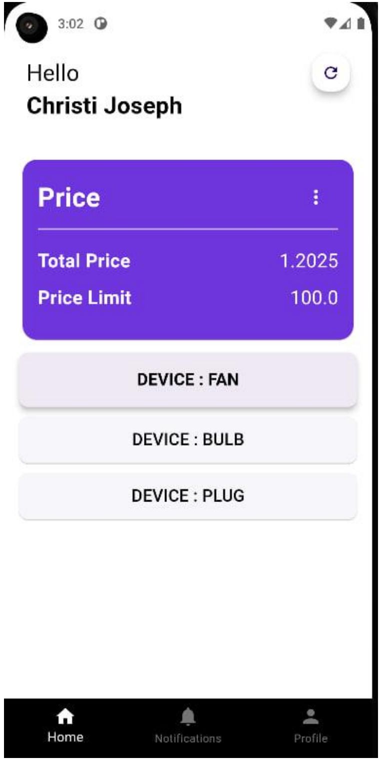

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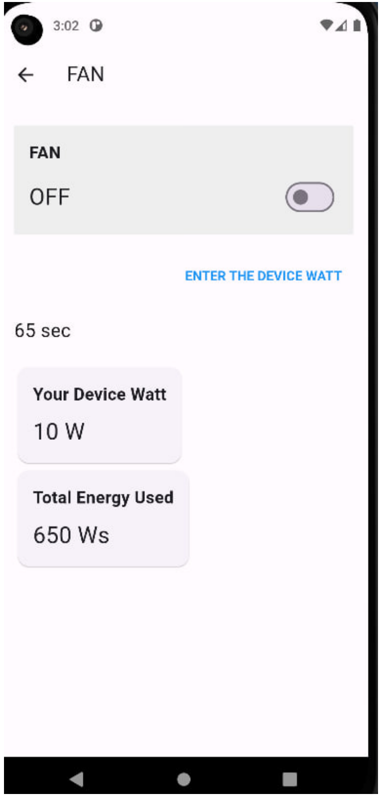
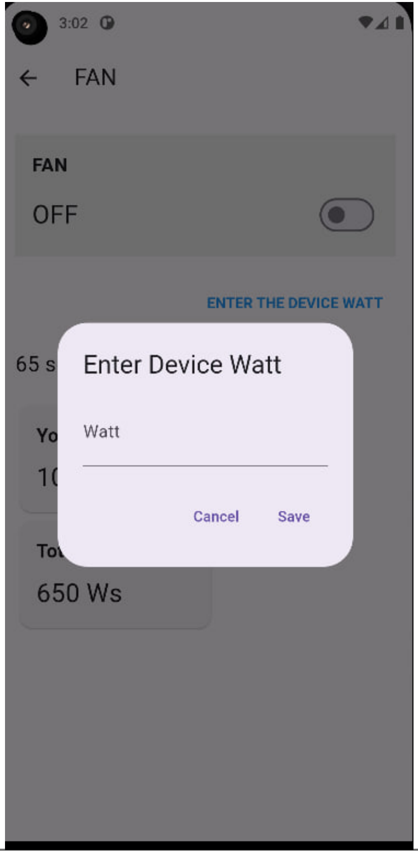
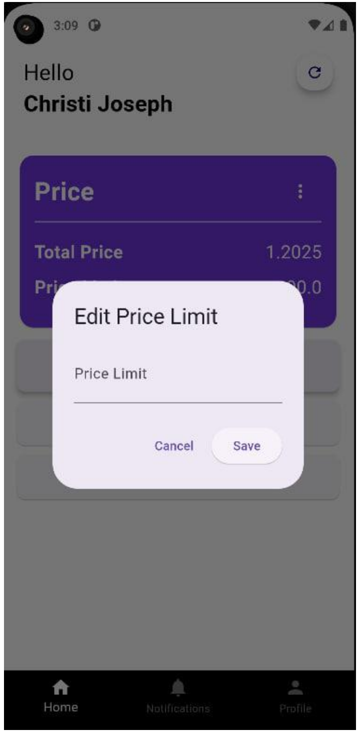
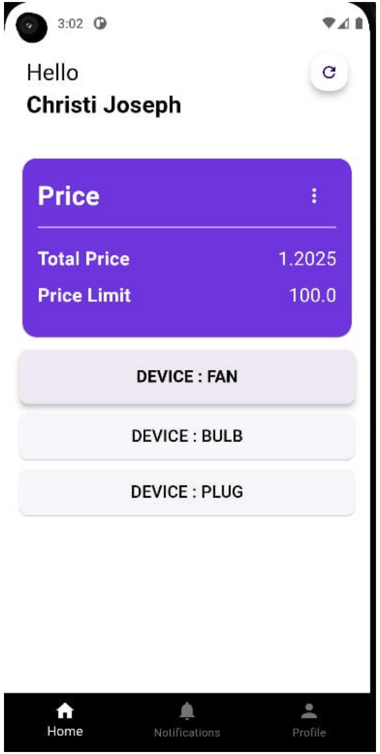


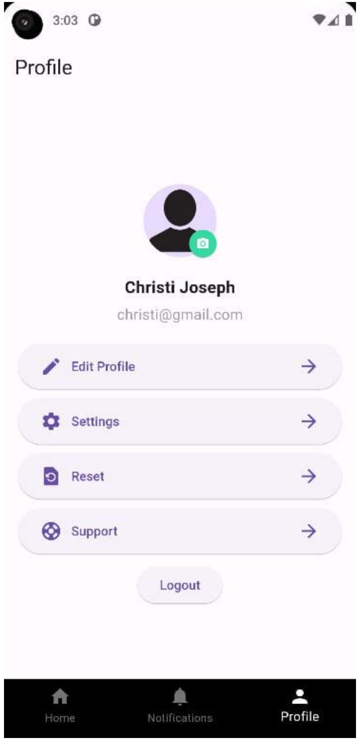


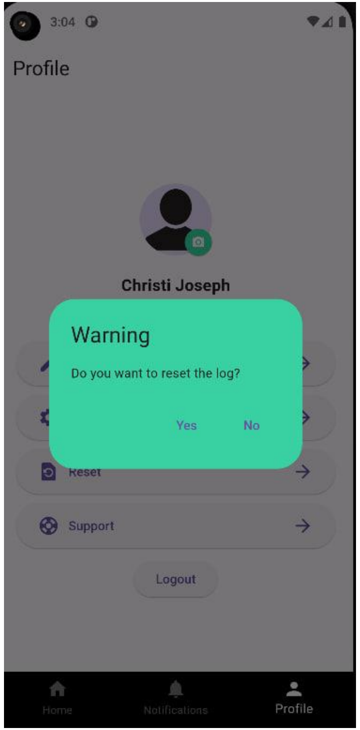
APPLICATION UI

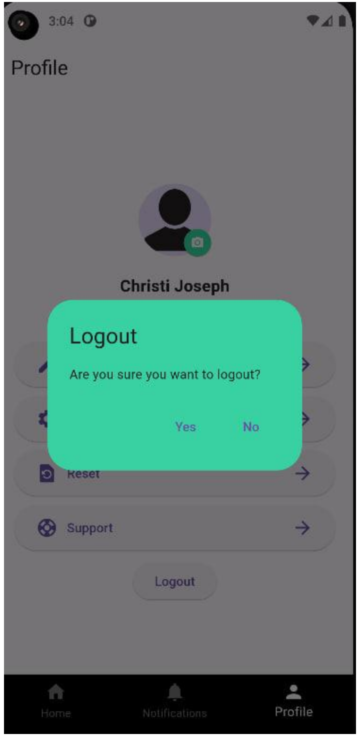




APPLICATION UI







Future Scope

•We look to add prediction module which gives usage recommendations to the user based on variation in the usage pattern of the user.

•Adding more devices points to the product in the future.

•Customizable product in which the user can change the number of devices according to their needs.

•An application that reduces the delay to negligable value.

CONCLUSION

The Smart Home Energy Management System (SHEMS) is a comprehensive solution for monitoring and managing household electricity consumption. It integrates current and voltage transformers for precise data collection, transmitted to a Firebase Firestore database via NodeMcu Esp8266 microcontroller. Real-time data is accessible through a mobile app, empowering users to make informed decisions and anticipate future expenses with predicted bill monitoring. Users can edit device wattage from the app, with corresponding energy calculations. A warning system alerts users when consumption exceeds set thresholds, and a PIR motion sensor detects room occupancy, suggesting powering down appliances in unoccupied rooms after an hour.

REFERENCES

[1] Ashton, Kevin, “That ‘internet of things’ thing,”, RFID journal, vol 22, no.7, pp. 97-114, 2009.

[2] Piyare, Rajeev, International Journal of Internet of Things, “Internet of things: ubiquitous home control and monitoring system using android based smart phone,” , vol. 2, no. 1, pp. 5-11, 2013.t

[3] J. K. Mishra, S. Goyal, V. A. Tikkiwal and A. Kumar ”An IoT Based Smart Energy Management System,” 2018 4th International Conference on Computing Communication and Automation (ICCCA) .

[4] Naeem Al-Oudat, Ahmad Aljaafreh, Ma’en Saleh, Murad Alaqtash, IoT-Based Home and Community Energy Management System in Jordan Procedia Computer Science, Volume 160, 2019

THANK YOU